LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Instrument elements and mechanism: - Elements, mechanisms, lever mechanisms, gears, hair springs, etc.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Instrument elements and mechanism: - Elements, mechanisms, lever mechanisms, gears, hair springs, etc.
5. Specific Objectives	15.1.1.2 Trainee will be able to explain the basic elements and mechanisms of an instrument.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of instrument mechanism assembly; Safety: Handle delicate springs with care; Supervision: Instructor monitors handling of hair springs and gears.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aircraft flight instrument?
(b) Name three basic elements found in a mechanical flight instrument.
(c) How does a change in physical quantity affect the instrument mechanism?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic instrument elements • Mechanism types overview • Lever mechanism principles • Gear train operation • Hair spring function	L & Ds	PPT, IB	15 min
(b)	• Detailed lever analysis • Gear ratio calculations • Hair spring torque characteristics • Mechanism assembly demonstration • Error sources in mechanisms	L & Ds	PPT, PDF	20 min
(c)	• Integration of elements • Mechanism reliability factors • Maintenance requirements • Troubleshooting common failures • Summary of key points	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a lever mechanism amplifies input motion.
(b) Describe the role of the hair spring in restoring the instrument pointer.
(c) Identify two factors that affect the accuracy of gear trains.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of lever mechanisms and gears will be applied in the next lesson regarding temperature compensation methods.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Temperature compensation of instrument mechanisms: - Bimetal strip method, thermo resistance method, thermo magnetic shunt method.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min